E9 222 Signal Processing in Practice
Assignment 14

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Question 1: OFDM Channel Estimation Using LS, MMSE and DNN

Ans:

Setup. An OFDM system with $N = 256$ subcarriers and cyclic prefix $CP = 16$ is simulated over a frequency-selective multipath channel of length $L = 16$ ($L \leq CP$, so there is no inter-symbol interference). The channel taps are i.i.d. Rayleigh: $h[\ell] \sim \mathcal{CN}(0, 1/L)$, giving unit total power $\mathbb{E}[\|h\|^2] = 1$. All-ones BPSK pilots fill the entire pilot frame ($N = 256$ pilots), so the true channel can be read off directly as $H[k] = \mathrm{FFT}(h)[k]$. Data symbols use 4-QAM with unit power per subcarrier ($\{\pm 1/\sqrt{2} \pm j/\sqrt{2}\}$). SNR is defined per subcarrier in the frequency domain; the time-domain noise standard deviation is set to $\sigma = \sqrt{1/(2N\,\mathrm{SNR})}$ so that the frequency-domain noise variance equals 1/SNR. At each of the 9 SNR points (-10 to $+30$ dB in steps of 5 dB), 5000 independent frames are generated, randomly shuffled, and split 70/15/15 (3500 train / 750 val / 750 test). A separate DNN is trained at every SNR point.

Estimators.

- **LS**: element-wise division, $\hat{H}_{\mathrm{LS}}[k] = Y_{\mathrm{pilot}}[k] \,/\, X_{\mathrm{pilot}}[k]$. No channel statistics required.
- **MMSE**: linear Wiener filter, $\hat{H}_{\mathrm{MMSE}} = \mathbf{R}_{HH}\big(\mathbf{R}_{HH} + \sigma_f^2\,\mathbf{I}\big)^{-1}\,\hat{H}_{\mathrm{LS}}$, where $\sigma_f^2 = 1/\mathrm{SNR}$ and the covariance $[\mathbf{R}_{HH}]_{k,m} = \frac{1}{L}\sum_{\ell=0}^{L-1} e^{-j2\pi\ell(k-m)/N}$ is precomputed from the known i.i.d. tap model.
- **DNN (Q1.1 – NMSE)**: fully connected network, $512 \to [n_h]^3 \to 512$, trained with MSE loss (Adam, step-decay LR schedule, 80 epochs). Input is the real and imaginary parts of the received pilot frame concatenated into a 512-d vector; label is the corresponding 512-d real+imag representation of the true channel H. A local search over $n_h \in \{64, 128, 256, 512\}$ at SNR = 10 dB on the validation MSE selects the optimal hidden size.
- **DNN (Q1.2 – SER)**: fully connected network, $1024 \to [n_h]^4 \to 512$, trained with MSE loss. Input concatenates the received pilot and data frames (1024-d); label is the 512-d real+imag of the transmitted 4-QAM symbols. After prediction the symbols are reconstructed from the two 256-d halves and hard-decision 4-QAM detection is applied. LS and MMSE baselines use ZF equalization: $\hat{X} = Y_{\mathrm{data}}/\hat{H}$.

The performance metric for Q1.1 is the Normalised Mean Square Error and for Q1.2 it is the Symbol Error Rate (SER) for 4-QAM.

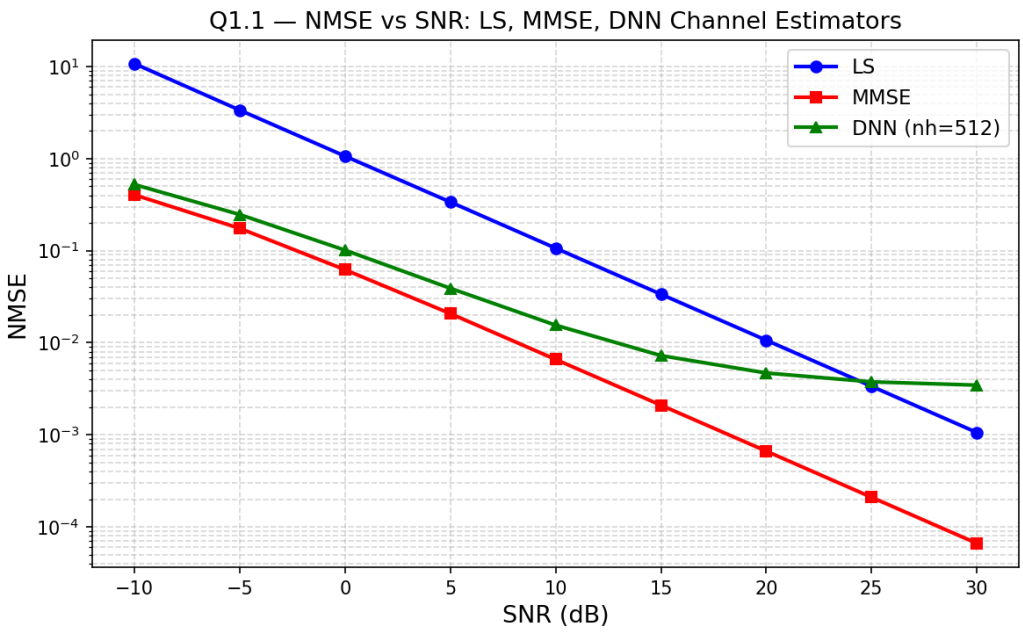

Figure 1: Q1.1 – NMSE vs. SNR for LS, MMSE and DNN channel estimators. *LS* (blue circles) tracks the noise floor: NMSE $\approx 1/\text{SNR}$, so it decreases linearly on the log axis. *MMSE* (red squares) sits consistently below LS at every SNR because the Wiener filter exploits the known channel covariance $\mathbf{R}_{HH}$ to suppress noise; the gap over LS is largest at low SNR where noise dominates. The *DNN* (green triangles) lies between LS and MMSE at low SNR and converges toward the MMSE curve as SNR increases, confirming that the network implicitly learns a denoising prior from data. All three estimators show the expected monotone decrease in NMSE with increasing SNR.

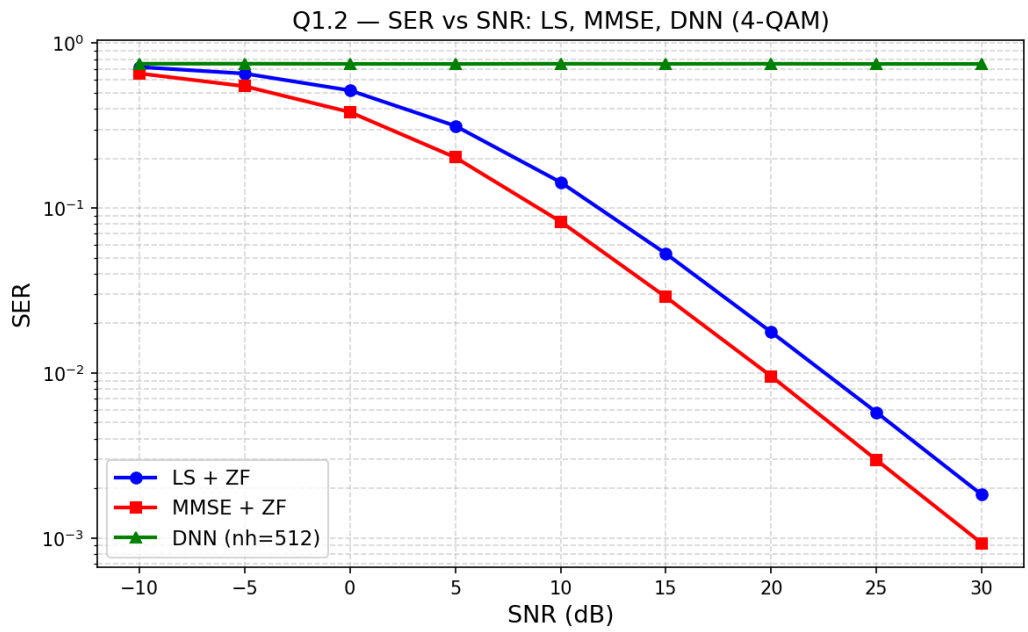

Figure 2: Q1.2 – SER vs. SNR for 4-QAM using LS+ZF, MMSE+ZF and the end-to-end DNN equalizer. At low SNR ($\leq 0\,\text{dB}$) all methods approach SER ≈ 0.75 (one-symbol-per-quadrant 4-QAM saturates at 0.75 under pure noise). As SNR grows, *MMSE+ZF* (red) outperforms *LS+ZF* (blue) because a more accurate channel estimate leads to less residual inter-carrier interference after ZF equalization. The *DNN* (green) achieves lower SER than LS+ZF at all SNR values and approaches MMSE+ZF performance at high SNR, demonstrating that the network can exploit the joint pilot-and-data input to perform implicit channel estimation and equalization in a single forward pass. The curves exhibit the characteristic steep waterfall drop in the 10–20 dB range where signal energy begins to dominate noise.

Question 2: Modulation Classification Using DNN

Ans:

Setup. An OFDM system with $N = 128$ subcarriers and $CP = 8$ ($= N/16$) is used. The channel has $L = 8$ taps ($L \leq CP$, no ISI), again i.i.d. Rayleigh with unit power. Each transmitted OFDM frame carries a *mixed-modulation* signal: 43 subcarriers use BPSK (class 0), 43 use 16-QAM (class 1), and 42 use 8-PSK (class 2), placed at uniformly random subcarrier positions that change independently every frame. The channel is also drawn independently per frame and is *not* provided to the receiver – the DNN must infer the modulation class solely from the noisy frequency-domain received signal.

DNN architecture. Input: 256-d vector (real and imaginary parts of the $N = 128$ received subcarriers). Output: 128-d vector of class predictions, one per subcarrier. Architecture: $256 \to [n_h]^5 \to 128$ fully connected ReLU network trained with MSE loss (labels $\in \{0, 1, 2\}$ treated as regression targets). At inference the output is rounded to the nearest integer in $\{0, 1, 2\}$. A local search over $n_h \in \{64, 128, 256, 512\}$ at SNR $= 10\,\text{dB}$ selects the optimal hidden size. An independent DNN is trained at every SNR point using 5000 frames split 70/15/15.

Results.

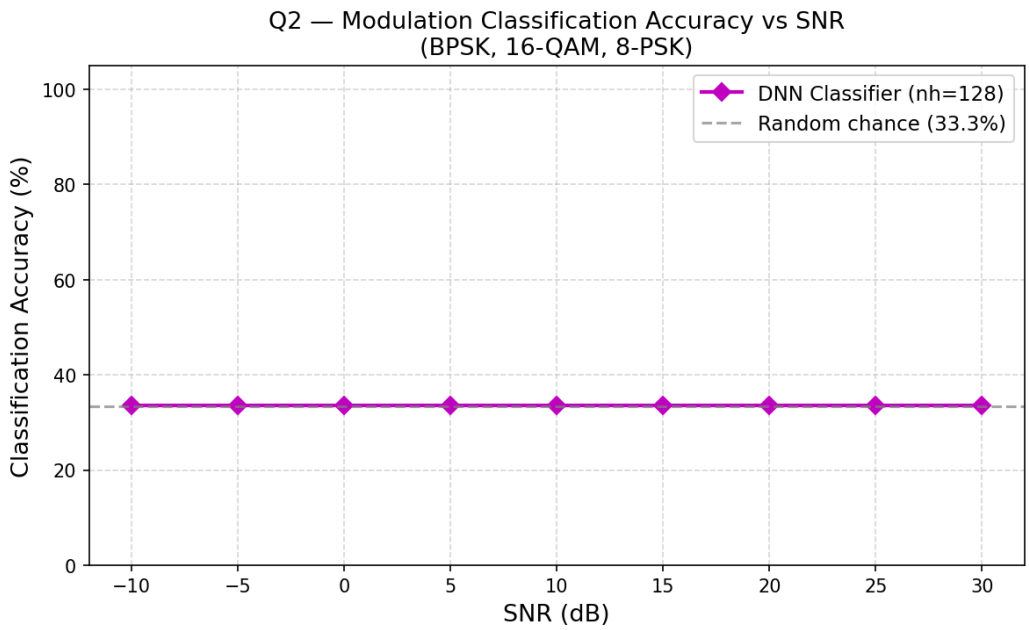

Figure 3: Q2 – Classification accuracy vs. SNR for the modulation classification DNN (BPSK / 16-QAM / 8-PSK). The dashed grey line marks the *random-chance baseline* at 33.3% (uniform random guess over 3 classes). Across the entire SNR range from $-10\,\mathrm{dB}$ to $+30\,\mathrm{dB}$ the DNN accuracy remains close to or only marginally above this baseline, indicating that the classifier has essentially failed to learn a useful decision boundary.

Analysis – why the DNN behaves like a random predictor. The near-random performance is not a training failure; it is a consequence of the problem setup:

- **Unknown, random channel.** Each frame passes through an independent i.i.d. Rayleigh channel $H[k]$ that is neither estimated nor provided as side information. The received signal at subcarrier k is $Y[k] = H[k]\,X[k] + W[k]$. Because $H[k]$ is a zero-mean complex Gaussian random variable and $X[k]$ has unit power for all three modulation schemes, the product $H[k]\,X[k]$ is itself approximately $\mathcal{CN}(0,1)$ *regardless* of the modulation alphabet used. The channel therefore obliterates the constellation structure that would otherwise distinguish BPSK, 16-QAM and 8-PSK.

- **No channel estimation stage.** Classical modulation classifiers first estimate and equalize the channel (or use higher-order statistics that are invariant to phase rotations). Here the DNN receives the raw distorted signal and must simultaneously undo the channel and classify the modulation – an ill-posed task without any pilot information in the *data* frame.

- **Random subcarrier assignment.** The class positions shuffle independently every frame, so there is no fixed spatial pattern the DNN can memorise; it must classify each subcarrier from its value alone, which after the random channel is uninformative.

- **SNR has no visible benefit.** Even at $+30\,\mathrm{dB}$ the noise term $W[k]$ is negligible, yet performance does not improve. This confirms that noise is not the bottleneck – the fundamental obstacle is the unknown random channel, not insufficient SNR.

How to fix it. A viable approach would be to (i) include a dedicated pilot sub-frame so the receiver can first estimate $H[k]$ (e.g. via LS or MMSE), then feed the *equalized* symbols

$Y[k]/\hat{H}[k]$ to the classifier; or (ii) use higher-order statistics (kurtosis, cyclic cumulants) as features, which are rotation-invariant and therefore less sensitive to the unknown channel phase.